

Intelligent Document Processing

A Tier-1 Indian bank went from ₹162 per supplier-invoice touch to ₹18. No new hires. Just a working IDP pipeline. Today we build our version.

How many invoices does your father's shop process monthly? How many hours does it take? That is the opportunity.

WHAT'S INSIDE

- 01** Why IDP, now
- 02** OCR options — pick by volume + accuracy...
- 03** Extraction — text → schema
- 04** Validation + HITL
- 05** 40-line reference pipeline

WHY THIS LESSON MATTERS

₹247 crore saved — by one invoice pipeline.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Describe the IDP pipeline end-to-end.

2

OCR a scanned PDF using Tesseract or Textract.

3

Extract structured fields via LLM prompting or spaCy NER.

4

Validate with business rules + confidence thresholds.

5

Route uncertain extractions to human review.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Why IDP, now

The stack has collapsed in 24 months

02

OCR options — pick by...

Tesseract for pilots. Cloud for production. LLMs when reasoning is needed.

03

Extraction — text → schema

Classical NER vs structured LLM prompting

04

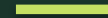
Validation + HITL

The non-negotiable last mile

01

PART 1 OF 4

Why IDP, now



The stack has collapsed in 24 months

CONCEPT 1 OF 4

Why IDP, now

Cloud OCR: 95%+ accuracy, ₹1-3 per page.
LLMs: field extraction in 40 lines of Python.
Every India bank / hospital / logistics player now invests.
The last-mile is ALWAYS human-in-the-loop.
STACK
OCR → LLM → Validate → HITL
Skip any one and you have a prototype, not a product.

DEFINITION

Why IDP, now

The stack has collapsed in 24 months

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

OCR options — pick by volume + accuracy need

Tesseract for pilots. Cloud for production. LLMs when reasoning is needed.

KEY POINTS

- Tesseract for pilots. Cloud for production. LLMs when reasoning is needed
- Practical, named, applied to a real Indian use-case.
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Extraction — text → schema

Classical (spaCy) — fast, deterministic, cheap.

LLM prompt — handles messy inputs, pay per call.

Prompt: "Extract as JSON: {invoice, supplier, total, line_items: [{sku, qty, price}]}".

json.loads the response; validate schema.

RULE

Classical if possible

LLM when inputs are too varied for rules.

DEFINITION

Extraction — text → schema

Classical NER vs structured LLM prompting

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Validation + HITL

Schema: did every required field come back?

Business rule: $\text{total} == \sum(\text{qty} \times \text{price})$?

Confidence: LLM meta-score OR OCR word-level.

Below threshold → human-review queue.

Humans correct; corrections feed back into few-shot.

QUEUE

Human in the loop

Every production IDP has this queue. Plan for it.

KEY POINTS

- Schema: did every required field come back?
- Business rule: $\text{total} == \sum(\text{qty} \times \text{price})$?
- Confidence: LLM meta-score OR OCR word-level
- Below threshold → human-review queue

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pick the OCR + extraction stack for 3 scenarios

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

Pick the OCR + extraction stack for 3 scenarios

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which step in the pipeline do you trust least?

Q2

Where would you place the human-review queue in your current workflow?

Q3

What would stop you from shipping a 40-line pipeline tomorrow?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Which step is ALWAYS mandatory in production IDP?

A

Tesseract

B

LLM

C

Human-in-the-loop

D

GPU

Which step is ALWAYS mandatory in production IDP?



CORRECT ANSWER

Human-in-the-loop

WHY

Every production pipeline has a human-review queue for low-confidence extractions

LLM extracted ₹12,99,000 at confidence 0.62. Action?

A

Auto-post

B

Retry

C

Route to human review

D

Reject

LLM extracted ₹12,99,000 at confidence 0.62. Action?



CORRECT ANSWER

Route to human review

WHY

Below threshold — route to human review. Do not auto-post

Business rule that catches a hallucinated total:

A

Total > 0

B

Total == $\Sigma(\text{qty} \cdot \text{price})$

C

Total is int

D

Total != null

Business rule that catches a hallucinated total:

C O R R E C T A N S W E R



Total == $\Sigma(\text{qty} \cdot \text{price})$

W H Y

Line items must reconcile with the total; mismatch → reject

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

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THE TAKEAWAY

"OCR reads. NLP understands. Rules protect. Humans close the loop. Skip any one and you have a prototype."

WHAT TO NOTICE

- 1 Why IDP, now
- 2 OCR options — pick by...
- 3 Extraction — text → schema
- 4 Validation + HITL

Mini assignment — OCR your utility bill

Take any Indian utility bill PDF you have. Build an end-to-end extractor.

DELIVERABLES

- OCR with Tesseract or a cloud API.
- Extract: amount, due date, account number.
- Validate with a simple business rule.
- Screenshot the output + commit script.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Screenshot + script URL on the forum.

ONE THING TO REMEMBER

“

"OCR reads. NLP understands. Rules protect. Humans close the loop. Skip any one and you have a prototype."

— AI Learning Hub

END OF LESSON 3

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Project 1 — Real Estate Price Prediction. Saturday's build-along.